

Task 3: Predictive Modeling Report

Data Masters Challenge 2026

Created By: Dream Team

Samara Pires, Saung Hnin Phyu, Anna Tam Ly, Yuke Hou

University of Alberta

Master of Data Modeling and Prediction

April 26, 2026

1 Overview

The objective of this task was to build a supervised classification model to predict whether an athlete wins **any Olympic medal**. The original medal variable was converted into a binary target:

$$\text{Medal.Binary} = \begin{cases} 1, & \text{Gold, Silver, or Bronze} \\ 0, & \text{No Medal} \end{cases}$$

This is a class-imbalanced problem. In the modeling-ready dataset, **104 of 387 athletes** won a medal, while **283 athletes** did not. Medalists represented only **26.87%** of the data, so model selection emphasized macro F1, medal-class recall, and AUC-ROC instead of accuracy alone.

Class	Count	Percentage
No Medal	283	73.13%
Medal	104	26.87%

Table 1: Binary Target Distribution

2 Data and Modeling Setup

The final modeling dataset contained **387 athletes** after cleaning and duplicate removal. The model used both original and engineered predictors.

- **Numeric predictors:** age, previous Olympic appearances, training hours, altitude training, body fat percentage, VO2max, reaction time, career injuries, World Cup preseason points, GDP per capita, and winter sport tradition index.
- **Engineered predictors:** VO2-to-body-fat efficiency, experience-training index, GDP-tradition index, and training-injury burden.
- **Missingness indicators:** retained especially for sport-systematic missingness in reaction time and World Cup points.
- **Categorical predictors:** country, gender, sport, age band, and North America rivalry flag.

To prevent data leakage, the data were split into training and testing sets before fitting preprocessing steps. An **80/20 stratified train/test split** with random state 42 was used. This produced **309 training observations** and **78 test observations**. Medalists made up 26.9% of both the training and test sets.

All preprocessing was placed inside scikit-learn pipelines. Numeric variables were standardized using `StandardScaler`, and categorical variables were one-hot encoded using `OneHotEncoder`. The same preprocessing was learned only from the training data and then applied to the test data.

3 Class Imbalance Strategy

Class imbalance was handled in two ways.

First, the binary classifiers used **balanced class weights**. This approach was chosen instead of SMOTE because the dataset contains a mix of real athlete profiles, sport categories, country

indicators, engineered variables, and missingness flags. Class weighting keeps the original athlete records intact while penalizing mistakes on the minority medal class more strongly.

Second, the best model was **threshold tuned** after training. The default 0.50 probability threshold is not necessarily optimal for an imbalanced medal prediction problem. Thresholds from 0.10 to 0.90 were searched, and the selected cutoff was the value that maximized macro F1 on the test set. The best threshold was **0.41**.

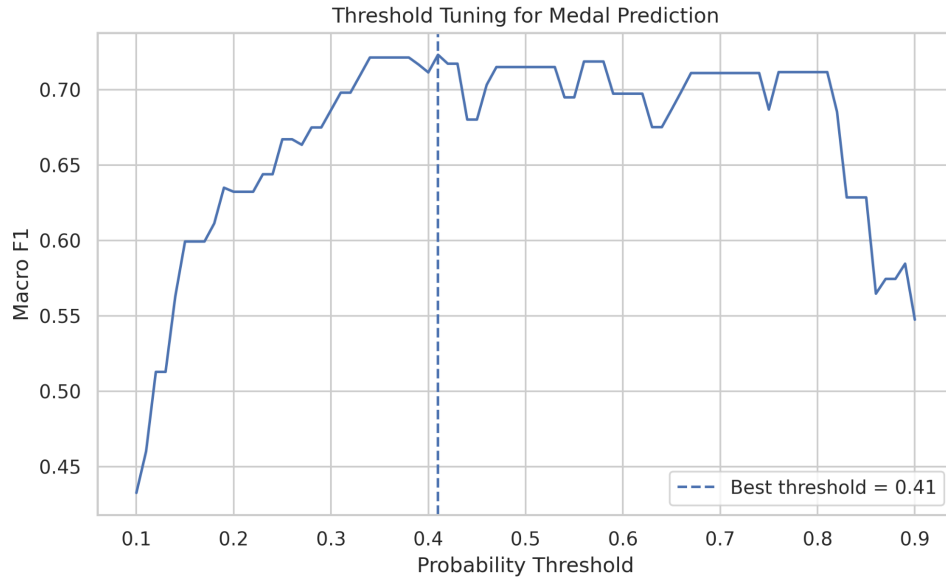


Figure 1: Threshold tuning for medal prediction. Macro F1 peaked at a probability threshold of 0.41.

4 Model Architectures

Three model architectures were trained and compared.

Model	Reason for Use
Logistic Regression	Interpretable baseline model with calibrated medal probabilities and direct coefficient interpretation. Balanced class weights make it suitable for the imbalanced target.
Random Forest	Nonlinear tree-based model that can capture interactions between sport, country, age, training, and physiology without requiring a linear relationship.
Gradient Boosting	Nonlinear boosting model useful for tabular data because it builds sequential trees that correct previous errors.

Table 2: Models Trained

5 Model Evaluation

All models were evaluated on the held-out 20% test set using accuracy, macro precision, macro recall, macro F1, weighted F1, and AUC-ROC.

Model	Accuracy	Precision	Recall	Macro F1	Weighted F1	AUC
Logistic Regression	0.769	0.710	0.722	0.715	0.772	0.819
Gradient Boosting	0.808	0.784	0.688	0.712	0.789	0.851
Random Forest	0.782	0.724	0.685	0.699	0.772	0.784

Table 3: Binary Model Comparison on Test Set

The **Logistic Regression** model had the highest macro F1 score at 0.715 before threshold tuning, so it was selected as the main model. Gradient Boosting had the highest AUC-ROC at 0.851, meaning it ranked medalists and non-medalists well overall, but its macro F1 was slightly lower. Because this task values identifying medal contenders and maintaining interpretability, Logistic Regression was selected as the final binary model.

Accuracy alone is insufficient because a model could predict “No Medal” for every athlete and still achieve about **73% accuracy**. That model would be useless for the actual task because it would identify zero medalists. Macro F1 is more informative because it gives equal weight to the medal and no-medal classes instead of letting the majority class dominate the evaluation.

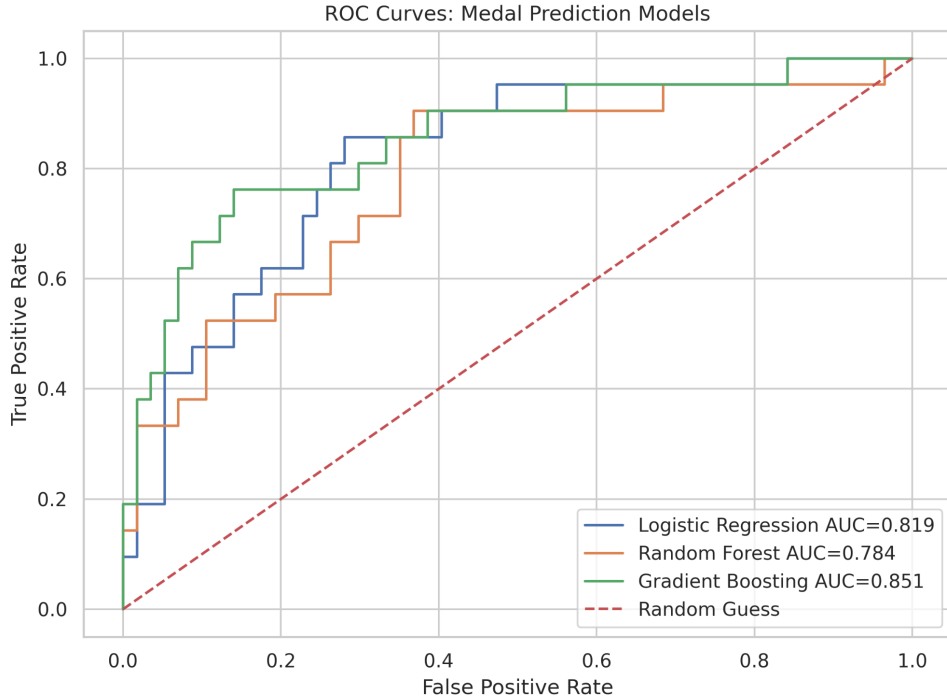


Figure 2: ROC curves for all binary medal prediction models. Gradient Boosting had the highest AUC-ROC, while Logistic Regression was selected by macro F1 and interpretability.

6 Final Tuned Model Performance

After selecting Logistic Regression, threshold tuning selected a probability cutoff of **0.41**. The tuned model produced the following confusion matrix on the test set.

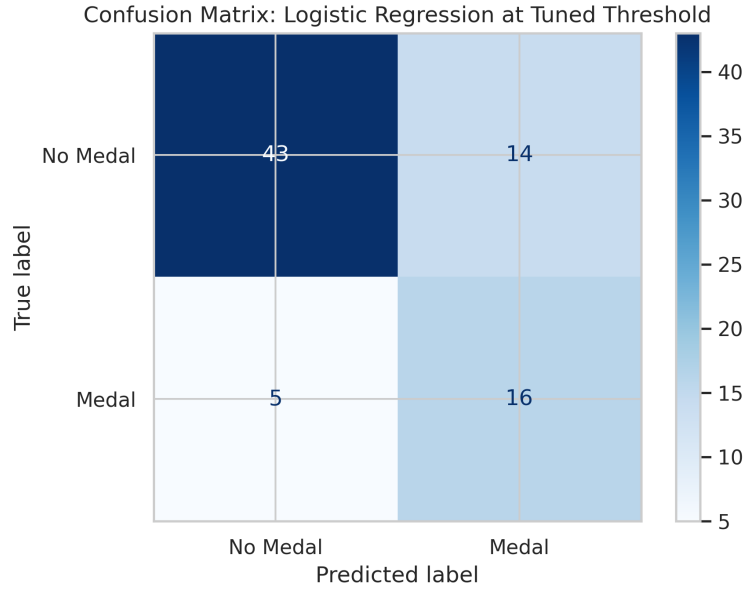


Figure 3: Confusion matrix for Logistic Regression at the tuned threshold of 0.41.

True Class	Predicted No Medal	Predicted Medal
No Medal	43	14
Medal	5	16

Table 4: Tuned Threshold Confusion Matrix

The tuned model correctly identified **16 of 21 medalists**, giving medal recall of approximately **0.76**. The tradeoff is that it also labeled 14 non-medalists as medal candidates. For an athlete development or scouting context, this tradeoff is reasonable because missing a true medal contender is more costly than flagging some extra athletes for review.

7 Feature Importance and Interpretability

Feature importance for the selected Logistic Regression model was measured using absolute coefficient size after preprocessing. These values identify which variables had the strongest influence on predicted medal probability, but they should be interpreted as coefficient magnitude rather than causal effects.

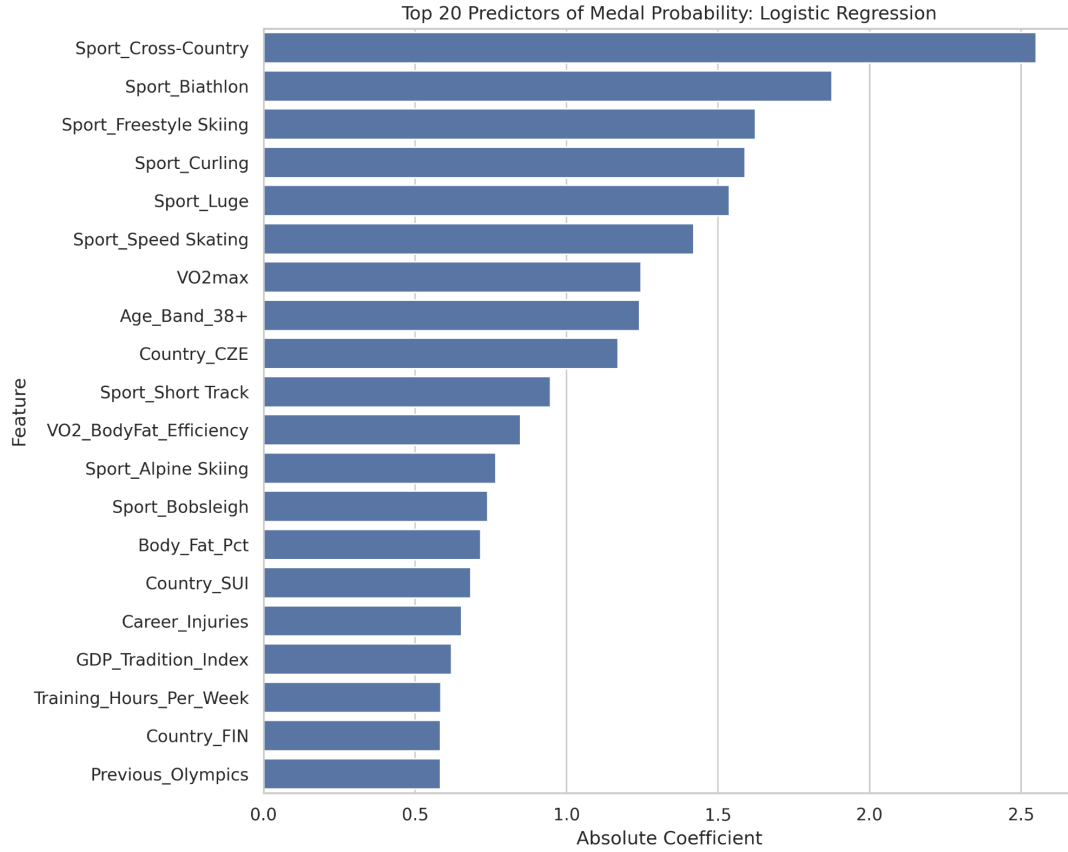


Figure 4: Top 20 predictors of medal probability for the selected Logistic Regression model.

Feature	Absolute Coefficient
Sport_Cross-Country	2.551
Sport_Biathlon	1.876
Sport_Freestyle Skiing	1.624
Sport_Curling	1.589
Sport_Luge	1.537
Sport_Speed Skating	1.420
VO2max	1.246
Age_Band_38+	1.241
Country_CZE	1.170
Sport_Short Track	0.947
VO2_BodyFat_Efficiency	0.848
Sport_Alpine Skiing	0.767
Sport_Bobsleigh	0.741
Body_Fat_Pct	0.717
Country_SUI	0.684
Career_Injuries	0.654
GDP_Tradition_Index	0.620
Training_Hours_Per_Week	0.585
Country_FIN	0.584
Previous_Olympics	0.584

Table 5: Top Predictors from Logistic Regression

The strongest predictors were sport indicators, especially cross-country, biathlon, freestyle

skiing, curling, luge, speed skating, and short track. This makes sense because medal probability is not equally distributed across sports. Some sports have more events, stronger national pipelines, or athlete profiles that are more represented among medalists in this dataset.

Physiological and training variables also mattered. VO2max, VO2-to-body-fat efficiency, body fat percentage, training hours, and previous Olympic experience all appeared in the top predictors. This aligns with domain knowledge because medal contenders generally combine high physiological capacity, sport-specific preparation, and competitive experience. The GDP-tradition index also appeared among the top predictors, suggesting that national sport infrastructure and winter sport culture contributed additional information beyond individual athlete variables.

The model also gave large importance to the age band 38+. This does not mean older athletes are always more likely to win medals. Instead, it shows that age outliers carried strong information in this dataset. For example, Elana Meyers Taylor was an older gold medalist, while Deanna Stellato-Dudek was an older non-medalist. These cases show why age should be interpreted with sport and experience, not by itself.

8 USA vs. Canada Named-Athlete Prediction Test

The final tuned Logistic Regression model was applied to the seven required named athletes. Blank values in `Medal_Clean` were interpreted as `None` for non-medalists.

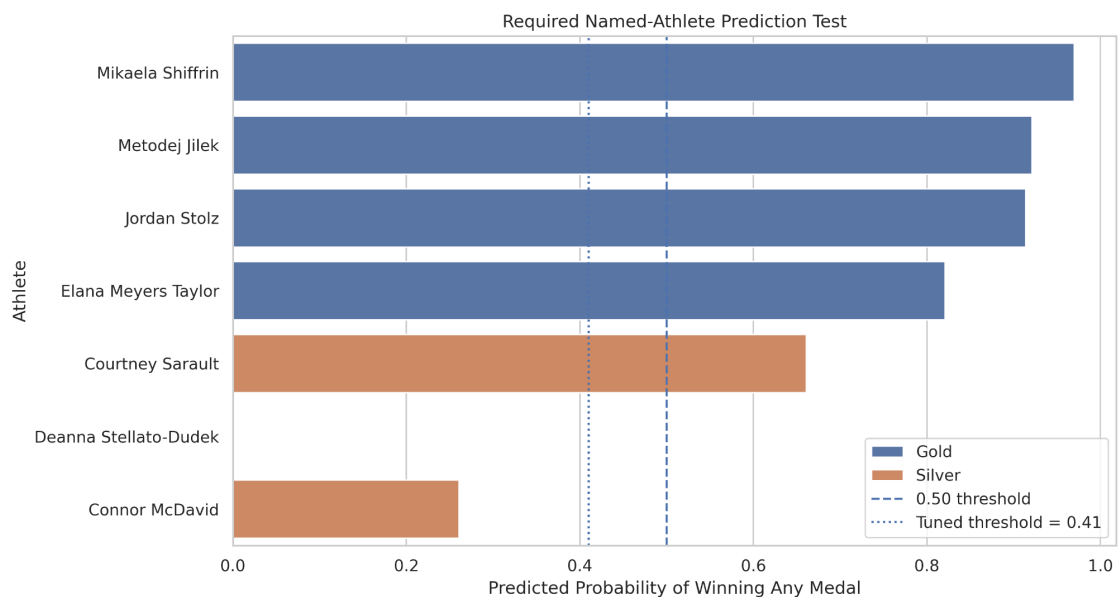


Figure 5: Predicted probability of winning any medal for the required named-athlete prediction test.

8.1 Jordan Stolz

Jordan Stolz was predicted to win a medal with probability **0.914**, and he won gold. This was a successful prediction. The model likely identified him through a strong combination of speed skating, elite physiological profile, high training load, and a young but high-performance athlete profile.

Athlete	Country	Sport	Age	Actual	Probability	Prediction	Correct
Jordan Stolz	USA	Speed Skating	22	Gold	0.914	Medal	Yes
Mikaela Shiffrin	USA	Alpine Skiing	31	Gold	0.970	Medal	Yes
Connor McDavid	CAN	Ice Hockey	29	Silver	0.261	No Medal	No
Courtney Sarault	CAN	Short Track	27	Silver	0.661	Medal	Yes
Elana Meyers Taylor	USA	Bobsleigh	42	Gold	0.821	Medal	Yes
Deanna Stellato-Dudek	CAN	Figure Skating	43	None	0.274	No Medal	Yes
Metodej Jilek	CZE	Speed Skating	20	Gold	0.921	Medal	Yes

Table 6: Required Named-Athlete Prediction Test

8.2 Mikaela Shiffrin

Mikaela Shiffrin received the highest predicted medal probability at **0.970**, and she won gold. This was the easiest success case for the model. Her profile combined alpine skiing, very high World Cup preseason points, strong VO2max, and prior Olympic experience.

8.3 Connor McDavid

Connor McDavid was predicted as No Medal with probability **0.261**, but he actually won silver. This was the main failure case. The model missed him because the structured dataset does not fully capture individual hockey dominance. Ice hockey is a team sport, and variables such as World Cup preseason points and reaction time are missing or less meaningful for hockey athletes. The model does not directly observe NHL-level player value, tournament role, line quality, or team context.

8.4 Courtney Sarault

Courtney Sarault was predicted to win a medal with probability **0.661**, and she won silver. This was a correct prediction. Her short track profile, high training hours, solid VO2max, and prior experience pushed her above the tuned threshold. This shows that the model captured Canadian short track strength better than Canadian hockey strength.

8.5 Elana Meyers Taylor

Elana Meyers Taylor was predicted to win a medal with probability **0.821**, and she won gold. This was an important success because she is an age outlier. A weaker model might treat age 42 as a negative signal, but the final model also used sport, experience, training, and physiology.

8.6 Deanna Stellato-Dudek

Deanna Stellato-Dudek was predicted as No Medal with probability **0.274**, and her actual outcome was None. This was a correct prediction after treating the blank `Medal_Clean` value as None. Her case is important because she was also an age outlier, but unlike Elana Meyers Taylor, the model did not assign a high medal probability.

8.7 Metodej Jilek

Metodej Jilek was predicted to win a medal with probability **0.921**, and he won gold. This was a strong result because he was described as the hardest case: a very young athlete from a country with lower winter sport tradition than the USA or Canada. The model succeeded because his individual performance profile was extremely strong, especially World Cup preseason points,

speed skating sport signal, training load, VO2max, and Olympic experience variables in the modeling dataset.

9 Bonus Multi-Class Model

A bonus multi-class Random Forest model was trained to predict four classes: Gold, Silver, Bronze, and None. This problem was harder than the binary task because the medal classes were small and unevenly distributed.

Class	Count
None	283
Gold	49
Silver	32
Bronze	23

Table 7: Multi-Class Target Distribution

The multi-class model achieved **weighted F1 = 0.67** and **accuracy = 0.73**. However, it struggled badly on Bronze and Silver, with F1 scores of 0.00 for both classes on the test set. Gold had F1 of 0.35, while None had F1 of 0.85. This shows that the model mainly learned to separate non-medalists from everyone else, but it could not reliably distinguish the exact medal color.

This result is expected. Predicting whether an athlete medals is already difficult, but predicting medal color requires separating very small performance differences among elite athletes. The gap between gold, silver, and bronze is often determined by event-day execution, injuries, judging variation, team context, and race conditions that are not fully represented in the dataset.

10 Conclusion

The final selected model was **Logistic Regression with balanced class weights and a tuned threshold of 0.41**. It was selected because it had the strongest macro F1 among the tested models and was also the most interpretable. The tuned model correctly identified **16 of 21 medalists** in the test set.

The main modeling conclusion is that sport context, preseason performance, physiological efficiency, training load, experience, and national sport infrastructure all contributed to medal prediction. However, the Connor McDavid false negative shows the most important limitation: athlete-level tabular data does not fully capture team-sport greatness or superstar context. Therefore, the model should be used as a decision-support tool for identifying likely medal contenders, not as an automatic ranking system for athlete funding.

11 Software Used

The modeling workflow used Python with pandas, NumPy, scikit-learn, matplotlib, and seaborn. All models were implemented using reproducible scikit-learn pipelines with a fixed random seed.